

Experiment 03: Interfacing DHT11 Sensor with Arduino UNO

Environmental Telemetry & Alert Actuation — Program Source Code & Experimental Output

1. Arduino C++ Source Code

Listing 3.1: Microcontroller Firmware for DHT11 Telemetry and Closed-Loop Alert Actuation.

```
1  /*
2  * Experiment 03: DHT11 Sensor Interfacing with Closed-Loop High-Temp Alert
3  * Pin Connections: DHT11 DATA -> Pin 7 | Alert Red LED -> Pin 8 | 5V & GND -> Power
4  * Alarm Trigger Threshold: 30.0 deg C (Adjustable setpoint)
5  */
6  #include <DHT.h>
7
8  #define DHTPIN 7           // DHT11 DATA connected to Digital Pin 7
9  #define DHTTYPE DHT11     // Sensor model designation
10 #define ALERT_PIN 8        // Red Alert LED connected to Digital Pin 8
11 #define TEMP_THRESHOLD 30.0 // Closed-loop alarm trigger setpoint in Celsius
12
13 DHT dht(DHTPIN, DHTTYPE);
14
15 void setup() {
16   Serial.begin(9600);
17   pinMode(ALERT_PIN, OUTPUT);
18   digitalWrite(ALERT_PIN, LOW); // Initialize alert LED to OFF
19   dht.begin();
20   Serial.println("=====");
21   Serial.println("  IoT Lab: DHT11 Telemetry & Alert System  ");
22   Serial.println("=====");
23   Serial.println("Time(s)\tHumidity(%) \tTemp(C)\tAlert Status");
24 }
25
26 void loop() {
27   delay(2000); // DHT11 1 Hz sampling rate constraint (2.0s polling interval)
28   float humidity = dht.readHumidity();
29   float temperature = dht.readTemperature();
30
31   // Validate sensor readout frame
32   if (isnan(humidity) || isnan(temperature)) {
33     Serial.println("ERROR: Failed to read from DHT11 sensor!");
34     return;
35   }
36   // Closed-Loop Threshold Comparison Logic
37   bool isAlert = (temperature >= TEMP_THRESHOLD);
38   digitalWrite(ALERT_PIN, isAlert ? HIGH : LOW);
39
40   // Stream formatted telemetry to Serial Monitor
41   Serial.print(millis() / 1000);
42   Serial.print(" s\t");
43   Serial.print(humidity, 1);
44   Serial.print(" %\t\t");
45   Serial.print(temperature, 1);
46   Serial.print(" C\t");
47   Serial.println(isAlert ? "ACTIVE [ALERT ON]" : "NORMAL [OFF]");
48 }
```

2. Experimental Output & Serial Monitor Telemetry

A. Temperature & Humidity Measurement Matrix:

#	Thermal Specimen State	Temperature	Humidity	Actuator Alert State
1	Ambient Laboratory Air	27.4°C	62.0%	OFF (Temperature Below Setpoint)
2	Ambient Laboratory Air	28.1°C	60.5%	OFF (Temperature Below Setpoint)
3	Heated Thermal Transit	31.8°C	74.0%	ON (High-Temp Alert Triggered)
4	Peak Heated State	34.0°C	86.0%	ON (High-Temp Alert Triggered)
5	Stabilized Baseline	28.5°C	65.0%	OFF (Normalized Below 30°C)

B. Serial Monitor Stream (9600 Baud):

Time(s)	Humidity(%)	Temp(C)	Alert Status
2 s	62.0 %	27.4 C	NORMAL [OFF]
4 s	60.5 %	28.1 C	NORMAL [OFF]
6 s	74.0 %	31.8 C	ACTIVE [ALERT ON]
8 s	86.0 %	34.0 C	ACTIVE [ALERT ON]
10 s	65.0 %	28.5 C	NORMAL [OFF]